produced very near the coalescing holes' horizons, they are made of the same material (a warpage of the fabric of spacetime) as the holes, they are not distorted at all by propagating through intervening matter—and, as a consequence, they can bring us detailed information about the holes and an unequivocal black-hole signature.

2. The ripples' symphony can tell us just how heavy each of the holes was, how fast they were spinning, the shape of their orbit (circular? elongated?), where the holes are on our sky, and how far they are from Earth.
3. The symphony will contain a partial map of the inspiraling holes' spacetime curvature. For the first time we will be able to test definitively general relativity's black-hole predictions: Does the symphony's map agree with Kerr's solution of the Einstein field equation (Chapter 7)? Does the map show space swirling near the spinning hole, as Kerr's solution demands? Does the amount of swirl agree with Kerr's solution? Does the way the swirl changes as one approaches the horizon agree with Kerr's solution?
4. The symphony will describe the merging of the two holes' horizons and the wild vibrations of the newly merged holes—merging and vibrations of which, today, we have only the vaguest understanding. We understand them only vaguely because they are governed by a feature of Einstein's general relativity laws that we comprehend only poorly: the laws' *nonlinearity* (Box 10.1). By "nonlinearity" is meant the propensity of strong curvature itself to produce more curvature, which in turn produces still more curvature—much like the growth of an avalanche, where a trickle of sliding snow pulls new snow into the flow, which in turn grabs more snow until an entire mountainside of snow is in motion. We understand this nonlinearity in a quiescent black hole; there it is responsible for holding the hole together; it is the hole's "glue." But we do not understand what the nonlinearity does, how it behaves, what its effects are, when the strong curvature is violently dynamical. The merger and vibration of two holes is a promising "laboratory" in which to seek such understanding. The understanding can come through hand-in-hand cooperation between experimental physicists who monitor the symphonic ripples from coalescing holes in the